Superpixel Consistency Saliency Map Generation for Weakly Supervised Semantic Segmentation of Remote Sensing Images

Xiaopeng Zeng, Tengfei Wang, Zhe Dong, Xiangrong Zhang, and Yanfeng Gu, *Senior Member, IEEE*

***Abstract*— The weakly supervised semantic segmentation (WSSS) method aims to assign semantic labels to each image pixel from weak (image-level) instead of strong (pixel-level) labels, which can greatly reduce human labor costs. However, there are some problems in WSSS of remote sensing images, such as how to locate labels accurately and how to get precise segmentation edges. To address these issues, we propose a novel framework directly transferring the scene classification model to perform semantic segmentation. We first train a multilabel scene classification network as the encoder to obtain the pretrained model, and then, the feature learned by the model is transferred to the decoder. Different from other methods, we propose a saliency map generator (SMG) instead of the class activation map (CAM) for more accurate location information by making pixels belonging to the same class lie close together while different classes are separated in feature space. Meanwhile, we take the superpixel patch as processing unit to provide precise boundary inhibition for the saliency map. To assign semantic labels for each patch, combined with extracted salient region, we propose a module responsible for exploiting the consistency of spatial and semantic similarity between different patches. Finally, we incorporate the above two modules to supervise the training process of the decoder without generating pseudolabels as most methods do, thus simplifying the training process. Experimental results show that our method outperforms other weakly supervised approaches on dense labeling remote sensing dataset (DLRSD) and Wuhan dense labeling dataset (WHDLD) with at least a 3% improvement on mean intersection over union (mIoU).**

***Index Terms*— Convolutional neural network (CNN), deep learning, semantic segmentation, weakly supervised learning (WSL).**

I. Introduction

With the rapid development of modern remote sensing technologies, semantic segmentation of remote sensing images has become an important topic of research in the field of remote sensing [1], [2]. In computer vision, semantic segmentation is a fundamental task that deals with assigning semantic labels (e.g., tree, building, and road) to each pixel in the image [3], [4]. Thus, semantic segmentation of remote sensing images enables land use and land cover (LULC) study on the earth's surface, facilitating a wide range of remote sensing applications, such as land resource mapping, natural environment monitoring, and autonomous agriculture [5]. The amount of remote sensing data is increasing sharply with the commercialization of satellites and the popularity of unmanned aerial vehicles, but traditional semantic segmentation requires pixel-level annotations, as illustrated in Fig. 1, each pixel needs to be given its semantic label in the training dataset. However, the manual annotation process is long and tedious, requiring high human labor and longtime costs, rendering it difficult to meet the needs of society. Therefore, weakly supervised learning (WSL) has become an emerging research hot spot in the field of machine learning [6], [7], which involves learning from weak supervision, such as image-level label, scribble, point, and bounding box, as opposed to strong labels used in fully supervised learning.

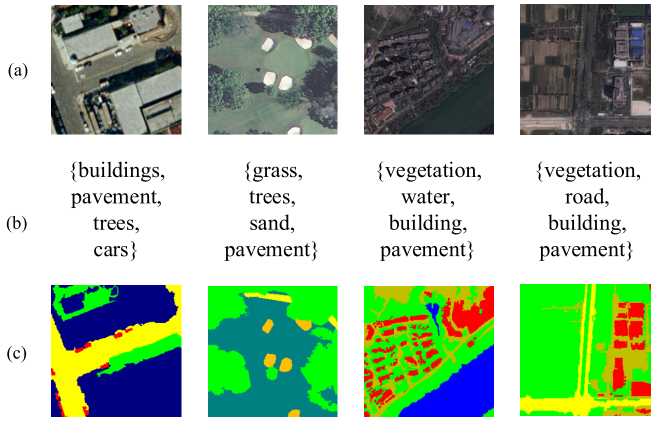


Fig. 1. Comparison of training labels for weakly and fully supervised semantic segmentation. (a) Image. (b) Weakly supervised labels. (c) Fully supervised labels.

Traditional methods for semantic segmentation include methods based on hand-craft features and methods based on machine learning. Methods based on hand-craft features [e.g., color histograms, texture features, and scale-invariant feature transform (SIFT)] strongly rely on relevant professional knowledge and experience. Differently, machine learning-based methods (e.g., support vector machine [8], K-means [9], random forest [10], and random fields [11]) have achieved

Manuscript received 23 September 2022; revised 31 January 2023 and 22 March 2023; accepted 25 March 2023. Date of publication 7 April 2023; date of current version 14 April 2023. This work was supported in part by the National Science Fund for Excellent Young Scholars under Grant 61522107 and in part by the Instance Segmentation and Analysis of Space Science Experiment Video. *(Corresponding author: Yanfeng Gu.)*

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Digital Object Identifier 10.1109/TGRS.2023.3264232

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better results compared with the former. However, it is difficult for these methods to establish complex function representation and so cannot adapt to semantic segmentation for complex remote sensing images [12]. In recent years, deep learning technologies represented by convolutional neural networks (CNNs) have made great progress and achievements in semantic segmentation [15]. Compared with traditional methods, CNNs have the ability to consider not only the features of the pixel itself but also the context of the pixel (i.e., the pixels near that pixel) in pixelwise classification [13], [14]. At present, most fully supervised semantic segmentation methods are developed based on the encoder–decoder structure of fully convolutional network (FCN) [16]. In addition, FCN and its variants, such as U-Net [17], SegNet [18], and DeepLab [19], [20], have been widely used in the remote sensing fields.

Most deep learning-based weakly supervised semantic segmentation (WSSS) methods adopt a multistage framework [21], [22], [23]. Particularly, these methods first train a classification model to provide weak location information and generate initial pseudolabels; then, the pseudolabels are refined and finally served as ground truth to train the segmentation model [24], [25], [26]. However, most WSSS works stick to natural scene image datasets (e.g., MS COCO and PASCAL VOC) instead of remote sensing datasets [27]. Moreover, WSL works in the field of remote sensing mostly pay attention to object detection [28], [29], [30] or localization [31], few focus on binary segmentation to extract ground object [32], such as road [33], [34], building [35], [36], and water [37], and even fewer centers on multiclass semantic segmentation [38], [39].

Although WSSS methods have achieved promising results for natural scene images, there are some challenges in WSSS of remote sensing datasets. The first challenge is that it is difficult for the existing works to locate different labels accurately in remote sensing images, which is caused by the characteristics of remote sensing images. Concretely, most methods employ the class activation map (CAM) [40] to get location information, but the CAM can merely obtain the region that has the greatest influence on the classification result in the image, which is suitable for the WSSS tasks in natural scene images. However, for remote sensing images, semantic classes may locate in any position of the image, and regions of some classes are irregular, so it is difficult to extract the most significant region correctly. Meanwhile, the CAM explores the high-level semantic information of the last convolutional layer but ignores the low-level features (e.g., texture, shape, and pattern), which play an important role in pixel-level classification for remote sensing images. As a result, the CAM cannot get ideal location information in remote sensing datasets. The second challenge is due to the label expansion methods adopted by the existing methods. They expand the initial labels through the different distributions of foreground and background. Different from natural scene images, there is no foreground or background for remote sensing images. Besides, the existing methods are difficult to obtain the category of pixels at object boundaries accurately [41]. Overall, the existing works have trouble getting precise segmentation

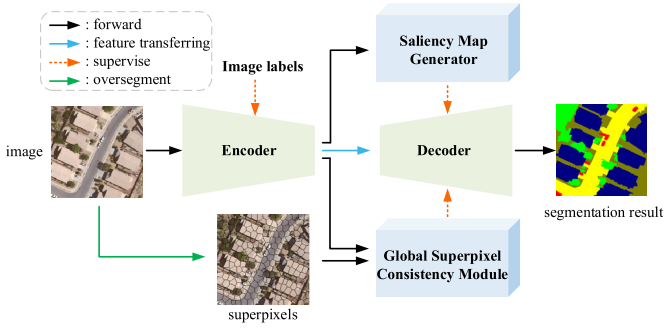


Fig. 2. Overview of the proposed framework.

edges for different classes [42], [43]. The third challenge comes from where most WSSS methods adopt a complex multistage framework. This paradigm needs to train multiple models and generate pseudolabels to train the segmentation network, which results in a complicated and time-consuming training process.

To address the above three challenges, we propose a framework to directly transfer the scene classification model to achieve semantic segmentation using image-level labels. The overview of the framework is shown in Fig. 2. First, we train a multilabel scene classification network to obtain the pretrained model. Then, we transfer the model as an input of the decoder used to upsampling for semantic output. To address the first challenge, we propose a module called saliency map generator (SMG) to extract the saliency map for more accurate location information. The SMG is an extension of our previous work [44], making the channel corresponding to the class with the largest probability of scene classification in the output results account for the maximum proportion of total output energy. This module encourages the network to map the pixels in feature space, so that pixels belonging to the same class lie close together, while different classes are separated by learning the feature for each semantic class of every image from the whole dataset. Consequently, it assigns semantic labels to the salient region of each semantic class appeared in the image. To tackle the second challenge, we propose the global superpixel consistency module (GSCM) to take the superpixel patch as the processing unit of the output map to improve the edge of the salient region for more precise edges in semantic segmentation results. Inspired by the human vision that objects of the same color are generally distinguished into the same class, the prior knowledge is that similar superpixel patches have the same semantic labels and vice versa in color space of the image. Besides, the GSCM employs the similarity measure to describe the global relationship across different superpixel patches and assigns semantic labels for all patches using the consistency of spatial and semantic similarity. For the third challenge, we simplify the training process by combining the above two modules to supervise the learning process of the decoder network without generating the pseudolabels. Our experiments show that the proposed method not only performs better than the CAM in extracting salient region but also exceeds the other weakly supervised methods by 3% for the mean intersection over union (mIoU).

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The main contributions of this article can be summarized as follows.

1) We propose a novel framework that can directly transfer the scene classification model to perform semantic segmentation using only image-level annotation, indicating that semantic segmentation is closely related to scene classification.
2) We propose a saliency map extraction method for remote sensing images, which can obtain more accurate location information than the CAM used by most WSSS methods.
3) We propose a global superpixel-based similarity measurement method to set precise edge constraints and assign semantic labels to each superpixel patch with the consistency of spatial similarity and semantic similarity.

The remainder of this article is organized as follows. Section II gives a brief review of the related works. Section III describes the framework of the proposed approach and related modules in detail. In Section IV, comprehensive experimental results and analysis are given. Discussions are given in Section V. Finally, Section VI draws the conclusion of the work.

II. Related Work

In this section, we briefly review the WSSS methods based on image-level labels and some weakly supervised deep learning works for the segmentation of remote sensing images. Recently, the WSSS has been rapidly gaining in popularity, with a spurt of research papers in the past few years. Generally, these methods usually fall into two categories: methods based on superpixel and methods based on the CAM, according to the inference method of label position.

Early works on the WSSS are based on superpixel oversegmentation [45], [46], they first oversegments the image to extract superpixel, then, the labels of superpixel patches are inferred through clustering or region merging, and the segmentation model is trained using the inferred pseudolabels finally. Typically, these methods extract the features of each superpixel, such as texture, color, shape, and SIFT, and use spectral clustering, multi-instance learning (MIL), or dictionary learning to infer the pseudolabels according to the similarity between the features. Besides, conditional random fields (CRFs) [47], Markov random fields (MRFs) [48], and CNN are commonly used as classification models. For example, Zhang et al. [49] used CNN and latent topic model to extract global and local features of the image, respectively, and combined CRF to learn the correlation between different labels. Lu et al. [50] cast the WSSS problem into a superpixel label noise reduction problem, and they formulated a sparse learning model to correct noisy labels. However, the superpixel-based methods are easily affected by the ability of superpixel feature representation, and the redundant pixel makes it difficult to infer the position of superpixel labels accurately.

Another line of research employs the CAM to acquire prior information. In the past few years, the CAM has demonstrated very promising performance in WSSS. These methods introduce the CAM to obtain the seed point for localization or initial pseudolabels and then adopt distance metric algorithms, such as region growing [51], random walk [52], and clustering [53] to get the refined pseudolabels as supervision to train the segmentation network. By following this line, there are two main research directions. Some researchers have tried to refine the CAM result to obtain more accurate initial labels by using dilated convolution [54], subcategory exploration [55], self-attention mechanism [56], adversarial climbing [57], and multiclass features [58]. Other researchers attempted to improve the seed expansion algorithms or the boundary of pseudolabels. In [51] and [59], they both set limits for initial seed points, growing trends, and expansion boundaries during the alternative training process. Moreover, Ahn and Kwak [52] proposed AffinityNet to learn the relationships between a pixel near the boundary and its neighborhood, and the random walk algorithm is performed to attain the refined pseudolabels. In addition to the above two lines, weak supervision information is utilized through MIL [60], expectation maximization [61], constrained CNN [62], saliency map [63], and others.

The aforementioned methods mostly focused on natural scene images, but the WSSS from remote sensing images is rarely explored. Also, most of these works focus on region extraction (i.e., binary segmentation), and even fewer concentrate on multiclass semantic segmentation. Yao et al. [38] used sparse autoencoders to solve the problem of ambiguous contents of the satellite images, and they aimed at LULC classification of tiles extracted from high-resolution images, so it is difficult to obtain the promising segmentation boundary for different regions in large-scale images. In [39], Nivaggioli and Randrianarivo introduced the method that is designed for natural scene images, but this method does not consider the characteristics of remote sensing images and contains complex steps, i.e., pseudolabel generation and refinement processes. Thus, there are still many challenges that should be addressed in more detail for weakly supervised deep learning-based semantic segmentation of remote sensing images [64], [65], [66].

III. Proposed Method

A. Encoder–Decoder Structure

We propose a novel weakly supervised deep learning framework for semantic segmentation of remote sensing images using image-level labels. As illustrated in Fig. 3, the main network of our method is in the form of an encoder–decoder structure, which is similar to FCN. We employ VGG16 [67] as the decoder of the framework, it consists of five downsampling layers, and each layer is made up of 3×3 convolutional layers, batch normalization (BN), rectified linear unit (ReLU) activation function, and pooling layer. Besides, the encoder is used to train a multilabel scene classification model. Then, the features extracted by the encoder are transferred to the decoder, which adopts the upsampling structure and fuses the information from low layer in the encoder by addition and convolution operation to attain segmentation results.

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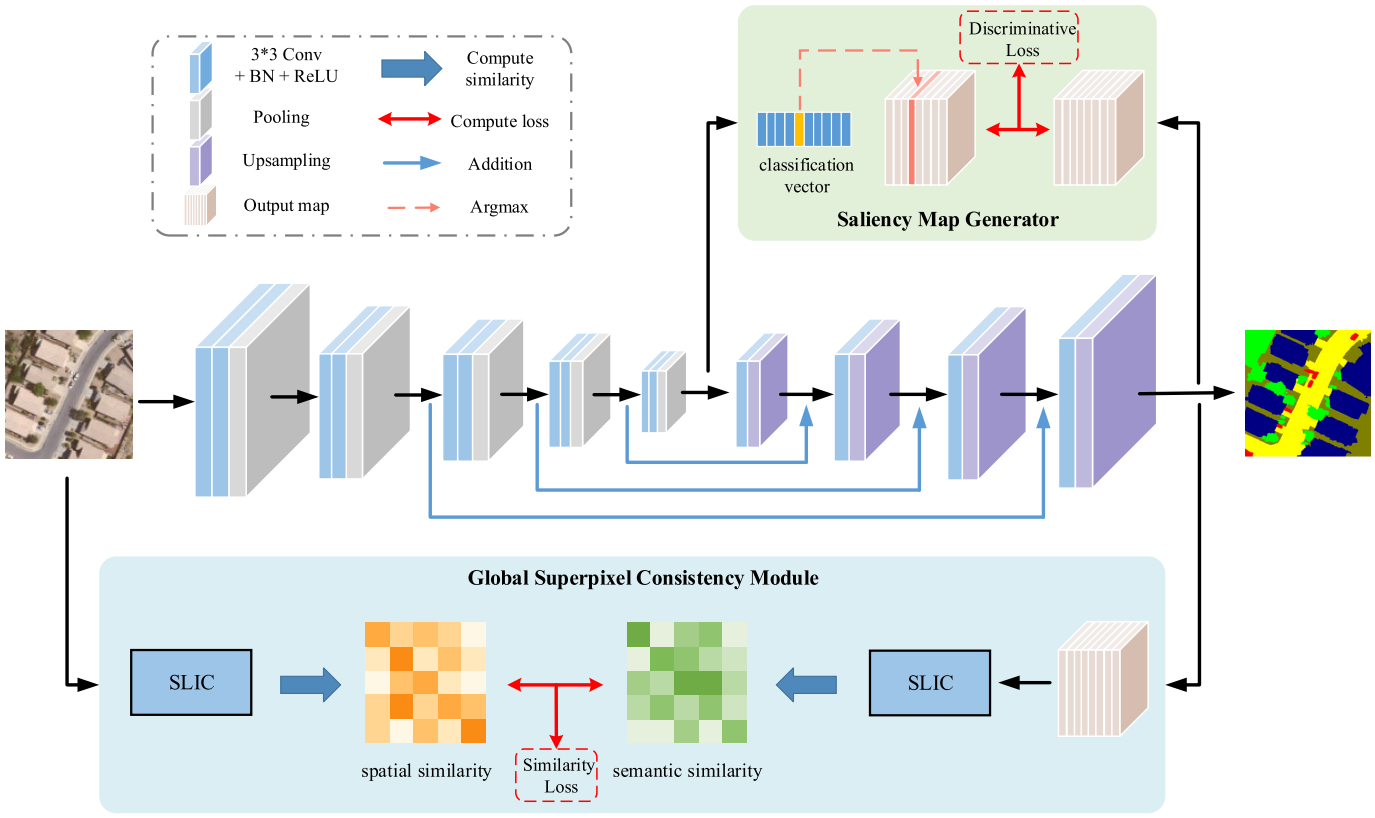


Fig. 3. Detailed architecture of the proposed framework. The framework consists of four components: the encoder, the decoder, SMG, and GSCM.

Also, we propose the SMG and the GSCM as supervisors to train the decoder by minimizing a joint loss function.

B. Saliency Map Generator

To achieve image classification, CNN extracts the feature of the most discriminative regions for the image, and semantic segmentation is essentially pixel-level image classification, so there may be some relationship between semantic segmentation and image classification in feature extraction and utilization. Driven by this insight, we propose a module called SMG to learn the features of each class from the whole training dataset and segment out the region corresponding to the most discriminative class in the image, establishing the relationship between semantic segmentation and scene classification.

We assume the training set $\mathcal{I} = \{I_1, \ldots, I_i, \ldots, I_M\}$, M is the number of training samples, and the size of each sample is $W \times H$, where W and H are the width and height of the image, respectively. The label of the sample is $y = \{y_1, \ldots, y_i, \ldots, y_M\}$, where $y_i = \{l_i^1, \ldots, l_i^c, \ldots, l_i^C\}$, and C is the number of categories. Besides, we define $l_i^c \in \{0, 1\}$ and set $l_i^c = 1$ when I_i contains the class c. In order to extract the salient region of the image, we first train a multilabel scene classification model by the encoder part of our framework. Moreover, the multilabel soft margin loss is a commonly used loss function in multilabel classification tasks [68], which combines binary cross entropy loss and sigmoid function to calculate the error between the prediction and the real label in logistic regression. Assuming the encoder is f, the multilabel soft margin loss is adopted to train f so as to make it learn the feature of different classes, and the loss is given by

$$\mathcal{L}_{\text{MLSM}} = -\frac{1}{C}\sum_{i=1}^{M} y_i \log(\sigma(v_i)) + (1 - y_i)\log(1 - \sigma(v_i)) \tag{1}$$

where $v_i = f(I_i)$ and v_i is the output vector consists of the classification probability of each class. In addition, σ denotes the sigmoid function and $\sigma(x) = 1/1 + e^{-x}$. Then, we let $\mathbf{X} = \{\mathbf{X}_1, \ldots, \mathbf{X}_i, \ldots, \mathbf{X}_M\} \in \mathbb{R}^{L \times H' \times W'}$ represent the feature extracted by f, where $H' = H/32$, $W' = W/32$, and L is the channel number of the output feature. We transfer $\mathbf{X}$ to the decoder g, and the output map is $\mathrm{Z} = g(\mathbf{X})$, where $\mathbf{Z} = \{\mathbf{Z}_1, \ldots, \mathbf{Z}_i, \ldots, \mathbf{Z}_M\} \in \mathbb{R}^{C \times H \times W}$. The most significant class of the image has the maximum value in the probability vector v_i, and it also owns the maximum response in the semantic output Z. Consequently, the class corresponding to the maximum value in v_i is considered as the most significant class $\hat{c}_i$ of the image, i.e., $\hat{c}_i = \arg\max(v_i) \in \{1, 2, \ldots, C\}$. Furthermore, we let the energy $\mathbf{Z}_i^{\hat{c}_i}$ of the $\hat{c}_i$th channel owns the largest proportion of the total energy in $\mathbf{Z}_i$ as the direction of gradient descent in the training stage. We define

$$E_i = \|\mathbf{Z}_i\|^2 = \frac{1}{C}\sum_{k=1}^{C} \left\|\mathbf{Z}_i^k\right\|^2 \tag{2}$$

as the energy of $\mathbf{Z}_i$; then, we yield the optimization object

$$\max \frac{E_i^{\hat{c}_i}}{E_i}, \quad \text{s.t. } \hat{c}_i = \arg\max(v_i) \tag{3}$$

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where $E_i^{\hat{c}_i}$ is the energy of the channel $\hat{c}_i$, and from (3), we can get

$$\max \frac{E_i^{\hat{c}_i}}{E_i} = \min \frac{\|\mathbf{Z}_i\|^2}{\left\|\mathbf{Z}_i^{\hat{c}_i}\right\|^2}. \tag{4}$$

To make (4) differentiable, we take the logarithm of it, and (4) becomes

$$\min \left(\ln \|\mathbf{Z}_i\|^2 - \ln \left\|\mathbf{Z}_i^{\hat{c}_i}\right\|^2 \right). \tag{5}$$

From (5), the discriminative loss $\mathcal{L}_d$ designed to extract salient region has the form

$$\mathcal{L}_d = \frac{1}{M} \sum_{i=1}^{M} \left(\ln \|\mathbf{Z}_i\|^2 - \ln \left\|\mathbf{Z}_i^{\hat{c}_i}\right\|^2 \right). \tag{6}$$

C. Global Superpixel Consistency Module

As mentioned above, the SMG could extract the most significant region in the image and assign a semantic label for it. However, SMG cannot guarantee the edge precision of different regions. Moreover, superpixel oversegmentation can preserve the boundary information of the object in the image. Therefore, in order to address the above drawback, the GSCM is proposed to assign the semantic label for all regions according to the intersuperpixel similarity and maintain good segmentation boundary of all ground regions in saliency maps.

First, we oversegment the input training set to generate superpixels $\mathcal{S} = \{S_1, \ldots, S_i, \ldots, S_M\}$ using the simple linear iterative clustering (SLIC) [69], and $S_i = \{s_1, \ldots, s_i, \ldots, s_N\}$ is the set of all superpixels in an image, where N denotes the number of superpixel patches and the index of each patch in S_i ranges from 1 to N. Also, to convert the processing unit from pixel to superpixel space, we average all pixels of each patch in the red, green, and blue (RGB) space, so that a single 3-D color value $C_i = (r_i, g_i, b_i)$ can represent the color features of one patch.

Then, we measure the global similarity of the image between different patches via the Euclidean distance metric. For the superpixel set S_i of the input image, the distance between s_i and s_j in S_i is $d_{ij} = \|C_i - C_j\|_2$, so the initial space similarity among them is defined as $\varphi_{ij} = 1 - d_{ij}$. However, the distance between different types of ground regions in RGB space is not far enough as we expect in remote sensing images. For that reason, we set a threshold T for φ_{ij} to enlarge the difference between different classes in RGB space, and we constrain φ_{ij} to 0 if $\varphi_{ij} < T$. The spatial similarity between different patches is

$$a_{ij} = \begin{cases} \varphi_{ij}, & \varphi_{ij} \geq T \\ 0, & \text{otherwise} \end{cases} \tag{7}$$

so we can get the global context spatial similarity matrix of S_i, i.e.,

$$\mathbf{A_S} = (a_{ij})_{N\times N}, \quad 1 \leq i \leq N, \ 1 \leq j \leq N \tag{8}$$

which represents the superpixel similarity of the input image.

Different from the spatial similarity computed from the input raw image I_i, the semantic similarity is obtained from the output semantic result $\mathbf{Z}_i$. Besides, $\mathbf{Z}_i$ is composed of $W \times H$ probability vectors of length C, and the vector denotes the classification probability of the pixel at the corresponding position in the image. Meanwhile, we only care about some classes which existed in the image I_i. Specifically, we select classes according to the given image-level label y_i in the training stage and set the threshold $Q = 0$ from the classification probability vector v_i to select classes that exist in the image. After selecting concerned classes, channels index to selected classes are chosen from $\mathbf{Z}_i$, while vectors of the unselected class are set to zero, and the chosen results are normalized by a softmax function finally.

Similar to the calculation process of the space similarity, we average the vectors belonging to the same superpixel index in the corresponding position of the image to get the result Z in superpixel space from Z_i, and $Z = \{z_1, \ldots, z_i, \ldots, z_N\}$ is with the size $N \times C$. A point that should be mentioned is that the index of superpixels in the input image I_i and the output $\mathbf{Z}_i$ shares the same parameters, i.e., the above two have the same index position of all superpixel patches for the same sample. Let b_{ij} be the similarity between different vectors in Z, we have

$$b_{ij} = 1 - \left\|z_i - z_j\right\|_2. \tag{9}$$

Furthermore, our goal aims at describing the global context relationship of all regions in the image; thus, we expect to obtain the similarity between one patch and any other patches. Consequently, we yield the global context semantic similarity matrix of the output Z_i

$$\mathbf{A_Z} = (b_{ij})_{N\times N}, \quad 1 \leq i \leq N, \ 1 \leq j \leq N. \tag{10}$$

Finally, we assume that regions with similar features in the color space have the same labels, i.e., the spatial similarity of input image is consistent with the semantic similarity of semantic result. By following this, the loss function $\mathcal{L}_s$ used to measure the consistency is:

$$\mathcal{L}_s = \frac{1}{M} \sum_{i=1}^{M} \left\|\mathbf{A_{Z_i}} - \mathbf{A_{S_i}}\right\|^2 \tag{11}$$

where $\mathbf{A_{S_i}}$ and $\mathbf{A_{Z_i}}$ are the spatial similarity and semantic similarity matrices obtained from the ith sample, respectively. We combine the above two loss functions to supervise the training process of the decoder in order to make the network learn from the prior knowledge as we expect. Overall, the joint loss function is

$$\mathcal{L} = \lambda \mathcal{L}_d + (1 - \lambda)\mathcal{L}_s \tag{12}$$

where λ is the loss coefficient to adjust and control the weight between the discriminative loss $\mathcal{L}_d$ and the similarity loss $\mathcal{L}_s$.

IV. Experiment and Analysis

A. Datasets and Evaluation Metrics

To verify the effectiveness of the proposed method, we performed the proposed method on two open datasets: dense labeling remote sensing dataset (DLRSD) [70], [71] and Wuhan dense labeling dataset (WHDLD) [72], for semantic

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TABLE I
DETAILS OF THE EXPERIMENTAL DATASETS

Dataset	Class number	Resolution(m)	Train samples	Test samples
DLRSD	17	0.3	1470	630
WHDLD	6	2.0	3458	1482

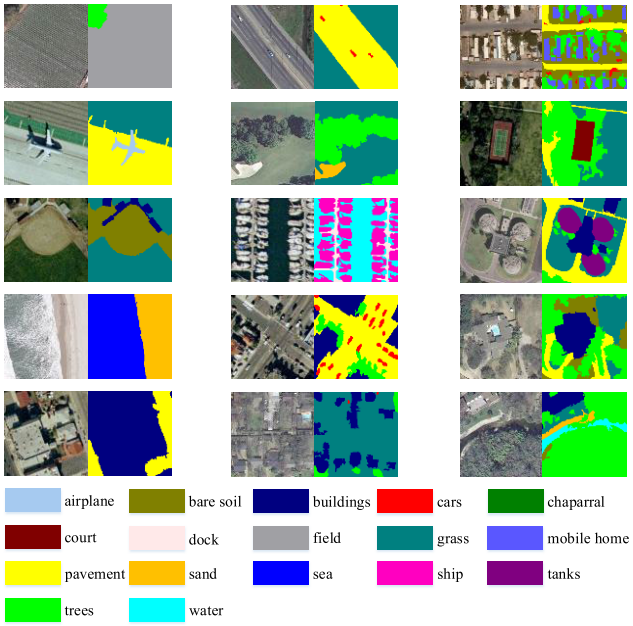


Fig. 4. Example images and corresponding labeling results in DLRSD dataset.

segmentation using image-level annotations. Both of these two datasets are from aerial orthoimagery in RGB form. Moreover, they are datasets with intensive manual pixel-level annotation, i.e., each pixel in the image is assigned a semantic label, so they can be used for multilabel classification and semantic segmentation (also known as classification in remote sensing). We randomly split datasets into train set and test set in a ratio of 7:3. The details of the datasets are listed in Table I.

Specifically, DLRSD dataset is an extension of UC Merced dataset derived from large remote sensing images in the United States Geological Survey (USGS) National Map of the USA regions, such as Birmingham, Boston, and Buffalo. Besides, UC Merced dataset consists of 21 LULC classes selected from aerial orthoimagery with a pixel resolution of one foot, containing 100 images of 256 × 256 size in each class. DLRSD densely tags the pixel of images in UC Merced with the 17 class labels, i.e., airplane, buildings, bare soil, chaparral, court, cars, dock, field, grass, mobile home, pavement, sand, sea, ship, trees, tanks, and water. Fig. 4 shows some images with corresponding pixel-wise labeling results, and the number of images present in the dataset for each class label is shown in Table II.

Besides, WHDLD dataset is the dataset composed of tiles extracted from large-scale remote sensing images of urban areas from Wuhan, China, with a total of 4940 images of 256 × 256 size in RGB form. Also, the pixels of each image are marked with the following six classification labels: bare soil, building, pavement, road, vegetation, and water. Fig. 5 shows some example images with the corresponding pixelwise

TABLE II
OVERVIEW OF DLRSD DATASET

Class	Number of Images	Label
airplane	100	1
bare soil	754	2
building	713	3
car	897	4
chaparral	116	5
court	105	6
dock	100	7
field	103	8
grass	977	9
mobile home	102	10
pavement	1331	11
sand	291	12
sea	101	13
ship	103	14
tank	100	15
tree	1021	16
water	208	17

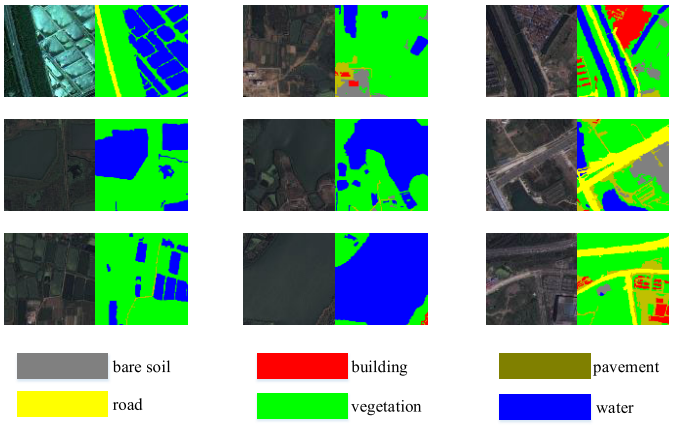


Fig. 5. Example images and corresponding labeling results in WHDLD dataset.

labeling results, and Table III shows the overview of WHDLD dataset including the contained six classes as well as the number of images and the label corresponding to each class.

We use overall accuracy (OA), average accuracy (AA), precision, recall, F_1 score, Kappa coefficient, user's accuracy (UA), producer's accuracy (PA), mIoU, and frequency weighted intersection over union (fwIoU) to evaluate the quantitative results of semantic segmentation comprehensively. The confusion matrix is widely introduced to analyze the result. OA is the accuracy of pixel-level classification, which is defined as the percentage of correctly predicted pixels in the total pixels. The calculation method is as follows:

$$\mathrm{OA} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{FP} + \mathrm{FN} + \mathrm{TN}} \tag{13}$$

where TP, FP, FN, and TN represent the true positive, false positive, false negative, and true negative, respectively. AA is the average pixel accuracy of all classes. Moreover, IoU is

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defined as the ratio of the intersection and union between the predicted result and ground truth. IoU is an indicator to describe the quality of the segmentation edge, and it is calculated by

$$\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}}. \tag{14}$$

mIoU is the mean of the IoU of all classes, and fwIoU weights the sum of the IoU of each class according to the frequency of occurrence of the class. Besides, the precision and recall are defined as TP/(TP + FP) and TP/(TP + FN), respectively, and the recall is the same as AA. Meanwhile, F_1 score is a harmonic average of precision rate and recall rate, with a maximum of 1 and a minimum of 0. F_1 score is defined as

$$F_1 = 2 \cdot \frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}}. \tag{15}$$

Kappa coefficient is an indicator used for consistency check concerning whether the model prediction results are consistent with the actual classification results. UA and PA are indicators evaluating the classification accuracy from the perspective of the user and producer, which are the same as the precision and the recall, respectively. We can get more reasonable performance results by a comprehensive comparison of different indicators.

B. Implementation Details

In GSCM, we employ SLIC to convert the processing unit from pixel to superpixel space. In addition, images sent to SLIC are cropped into 256 × 256 size, the smoothness of the superpixel is set to ten, and the number of iterations is ten. For DLRSD dataset, the region size of the superpixel is set to 16, and we set it to 12 for WHDLD dataset due to the low spatial resolution of images. After extracting the superpixel patches, we divided the values on the three channels of RGB space by 255 for normalization. In order to enlarge the difference of similarity between different superpixel patches, we set thresholds $T = 0.9$ in both two datasets. In terms of training strategy, in order to improve the stability of training process and make the network converge toward the desired direction, we froze the parameters of the encoder when training the decoder. Meanwhile, for the setting of λ, we adopt the form of exponential attenuation, i.e., $\lambda = \alpha e^{-i/\beta}$, where α is the initial weight of $\mathcal{L}_d$, β is the attenuation coefficient to adjust the ratio of two losses, and i is the number of iterations of training phase. In the early stage of training, semantic labels are first assigned to salient regions, while in the later stage, we gradually reduce the proportion of $\mathcal{L}_d$ (i.e., the value of λ) and increased the proportion of $\mathcal{L}_s$, in order to assign semantic labels to other regions and maintain the edge of different regions. We set $\alpha = 0.5$ and $\beta = 10$ for DLRSD dataset but set $\alpha = 0.6$ and $\beta = 10$ for WHDLD dataset, because the scene in WHDLD is more complex, making it more difficult to extract the similarity and distinguish different types of scenes. Therefore, we set a larger value of α on WHDLD dataset, and the value of β is the same as the former.

During the training phase, we set the total epoch to 15, the batch size is 128, and the initial learning rate is 0.1. Besides, the learning rate is updated and adjusted by step strategy according to the number of training iterations and it decreases by half every ten epochs. We employ SGD as an optimizer and use weight decay, which is set to 0.0005, and the momentum is 0.9. The implementation is based on PyTorch open-source framework and is training on two NVIDIA Tesla V100 GPUs with 32-GB memory.

C. Experimental Results

We apply the proposed SMG to DLRSD and WHDLD datasets to extract the salient region of the image, and the results of SMG are compared with CAM. Results of the saliency map from SMG and CAM on two datasets are shown in Fig. 6. The response heatmap of the salient region extracted by our method is better than CAM in terms of edge and location accuracy, which can provide better location information. CAM only multiplies the output feature map of the last convolutional layer of CNN and the parameters of the fully connected layer and then sums on the corresponding channel to obtain the response of every pixel in the feature map for each class. Compared with CAM, SMG adopts the cascading structure of downsampling (encoder) and upsampling (decoder), combines both high and low layer features, obtains saliency map from the output feature map of the decoder, and learns the features of each class from the whole training dataset, thus is superior to CAM in feature extraction and representation.

To demonstrate the superiority of our method, we compare our proposed method with other WSSS methods. The following six methods are considered for comparison.

1) MIL-FCN: Multiclass MIL loss is used to constrain the maximum value of the output segmentation map, and the response of the class is constrained by controlling the maximum value ranging from 0 to 1 [60].
2) Grad-CAM: The parameters obtained by the backpropagation of the full connection layer are weighted and summed with the feature map of the last convolution layer; then, the output is threshed into pseudolabels to train the segmentation network [73].
3) SEC: Loss functions for seed, expand, and constrain are designed to restrict the expansion of initial location information obtained by CAM during the training phase, and the pseudomasks are obtained to train the segmentation network [59].
4) DSRG: The most discriminative small region is used as the initial seed, and then, the seed region growing (SRG) algorithm is utilized and improved to expand the region of labels from the seed region; thus, the supervised regions are obtained through iteration in the training process [51].
5) PSA: The coarse pseudomask obtained from CAM is refined by AffinityNet and random walk based on the distance metric, and the obtained results are used as annotations to train the segmentation network [52].

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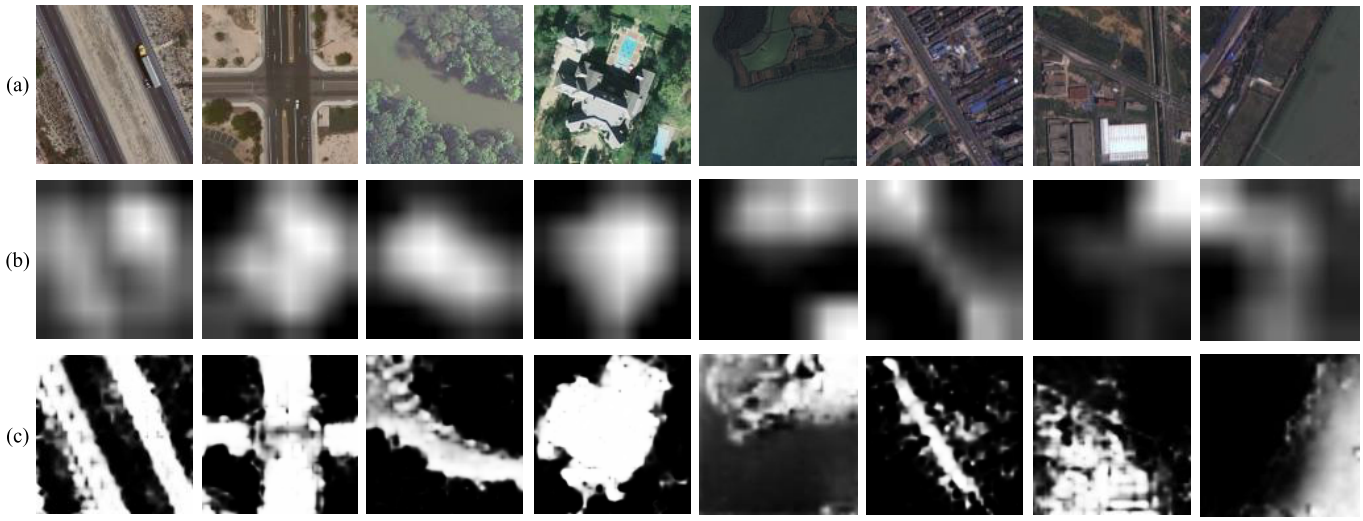


Fig. 6. Results of saliency map extracted by the CAM and our proposed SMG on DLRSD and WHDLD datasets. (a) Image. (b) CAM. (c) SMG.

TABLE III
OVERVIEW OF WHDLD DATASET

Class	Number of Images	Label
building	3722	1
road	3162	2
pavement	3881	3
vegetation	4631	4
bare soil	3539	5
water	3886	6

6) SEAM: Self-attention is used to attain a better result from CAM, and the gap between full supervision and weak supervision can be reduced by the regularization method [56].

In Table IV, we report the segmentation performance of the proposed approach and compare it with other approaches. The best scores are shown in bold in the table. Besides, Figs. 7 and 8 illustrate the visual performance of all approaches. Experiments demonstrate that the proposed method outperforms other listed methods. In terms of quantitative results, in DLRSD dataset, our method can reach 62.43% and 45.48% in recall and mIoU, respectively, and both of them are at least 5% higher than other methods. For WHDLD dataset, our method can reach 33.11% in mIoU, which is at least 3% higher than other methods, but it is not so good as Grad-CAM in recall with only 0.42% difference. Besides, from the visualization results of the two datasets, compared with other methods, our proposed method not only obtains more precise location for each class but also has better edge of ground regions. In Tables V and VI, we list the IoU of all classes in DLRSD dataset, as well as IOU and OA in WHDLD dataset. The IoU of "dock" is zero in DLRSD, and PA and IoU of "pavement" are zero in WHDLD, because the region of the above two classes belongs to fewer samples with small shape and tiny region in the dataset, indicating that our method is difficult to learn the feature of dim and small objects.

D. Ablation Study

The method proposed in this article mainly includes two modules, i.e., SMG and GSCM, used as supervisors in training process. To prove the effectiveness of each module and analyze the impact of each module on the result, we performed the ablation experiments on two datasets. We apply SMG and GSCM as loss $\mathcal{L}_d$ and $\mathcal{L}_s$ to supervise the proposed framework, respectively. Table VII shows the experimental results, and the results of the two datasets are visualized in Figs. 9 and 10.

Experimental results show that SMG can obtain rough segmentation results, extract salient regions, and assign semantic labels to them but cannot make the ground object have accurate enough boundary. In addition, when the GSCM acts alone, it cannot control the learning direction of the model in training process, and the segmentation results are poor without any constraint for significant classes. Therefore, SMG is needed to constrain the training process by learning the features of different classes from the distribution of the whole training set. On the whole, SMG can make sure that salient regions are correctly classified, and the edge of different regions will be improved when combined with the similarity between the superpixel patches in the GSCM.

E. Computational Efficiency

To compare and evaluate the computational efficiency of different methods, we conduct additional experiments to count and compare the time–cost, including training time and testing time. In our experiments, training time refers to the period of time from the beginning of training to the time when the value of the loss function is no longer decreasing. Testing time is the time it takes to complete the inference for all test samples using the trained model. Besides, we employ frames per second (FPS), which is the number of images processed per second in the testing stage to measure the inference speed of the method.

The results in Table VIII tell that MIL-FCN, Grad-CAM, SEC, and DSRG these four methods require shorter train-

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TABLE IV
EXPERIMENTAL RESULTS (%) OF DIFFERENT WSSS METHODS ON TWO DATASETS

Dataset	Method	OA	recall	precision	F_1	Kappa	mIoU	fwIoU
DLRSD	MIL-FCN [60]	23.76	9.47	9.56	9.14	9.36	4.64	11.76
	Grad-CAM [73]	52.18	55.55	53.99	46.54	45.16	33.31	34.84
	SEC [59]	54.59	51.52	48.03	48.43	48.30	33.05	40.25
	DSRG [51]	58.96	50.24	56.07	48.44	51.70	33.94	41.48
	PSA [52]	64.19	52.70	57.71	53.08	58.71	39.40	49.24
	SEAM [56]	63.18	56.36	56.23	54.64	58.46	39.98	49.01
	proposed	**64.28**	**62.43**	**58.71**	**63.07**	**60.09**	**45.58**	**50.42**
WHDLD	MIL-FCN [60]	22.64	16.75	19.38	16.64	9.87	9.66	15.13
	Grad-CAM [73]	40.53	**49.31**	40.53	36.74	27.52	23.69	20.68
	SEC [59]	58.94	35.38	42.84	41.18	36.17	26.56	42.64
	DSRG [51]	55.84	40.00	43.96	42.45	35.52	28.80	40.97
	PSA [52]	54.14	45.85	44.59	44.18	36.46	29.83	37.04
	SEAM [56]	49.90	30.48	39.88	34.55	12.53	14.85	15.32
	proposed	**63.99**	48.89	**49.12**	**52.86**	**49.48**	**33.11**	**50.02**

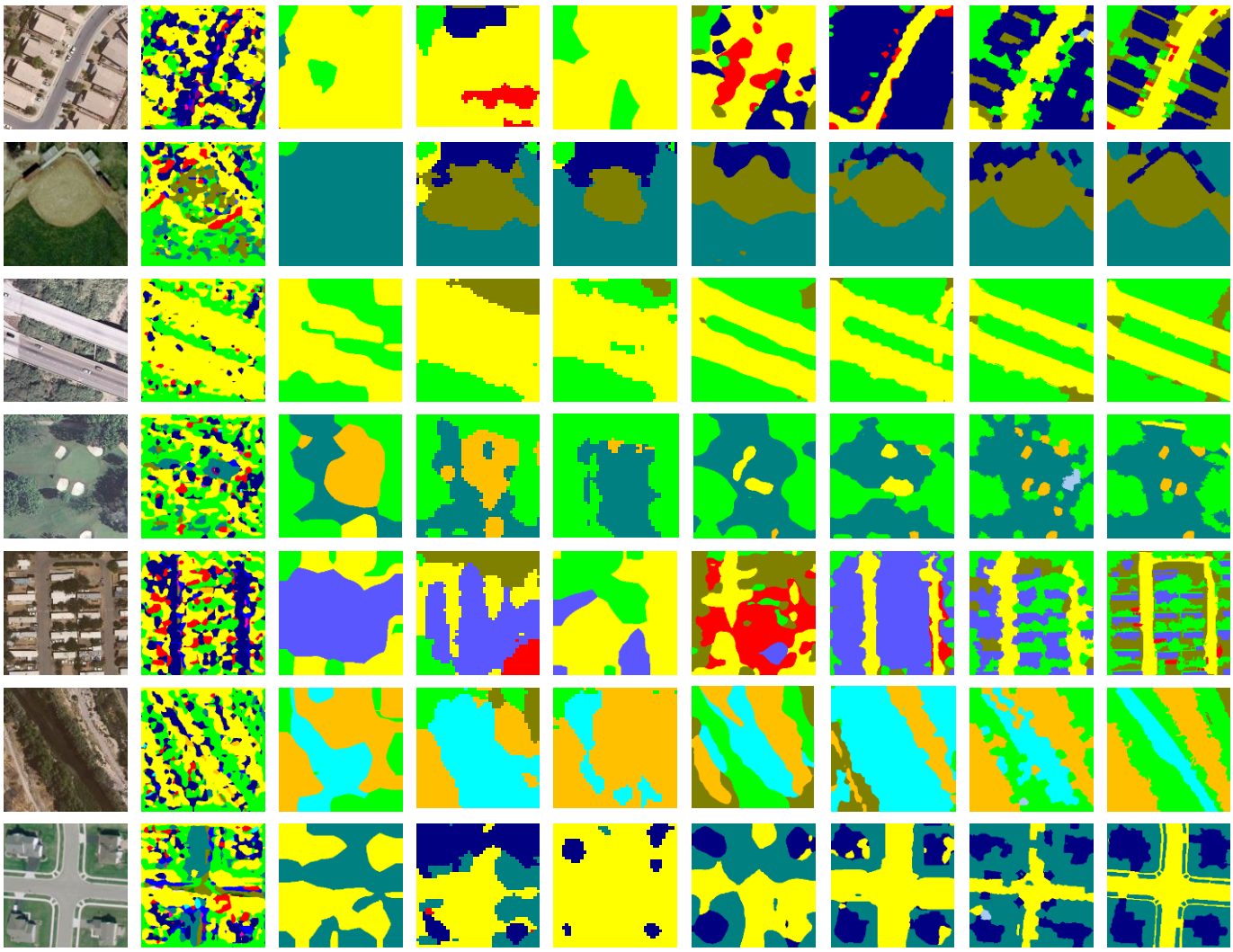


Fig. 7. Qualitative results of WSSS using different methods on DLRSD dataset. (a) Image. (b) MIL-FCN. (c) Grad-CAM. (d) SEC. (e) DSRG. (f) PSA. (g) SEAM. (h) Proposed. (i) Ground truth.

ing and testing time, but their quantitative performance and visualization results are too poor to be put into practice. Although PSA and SEAM can achieve better results than the above four methods, they take too much time during the test

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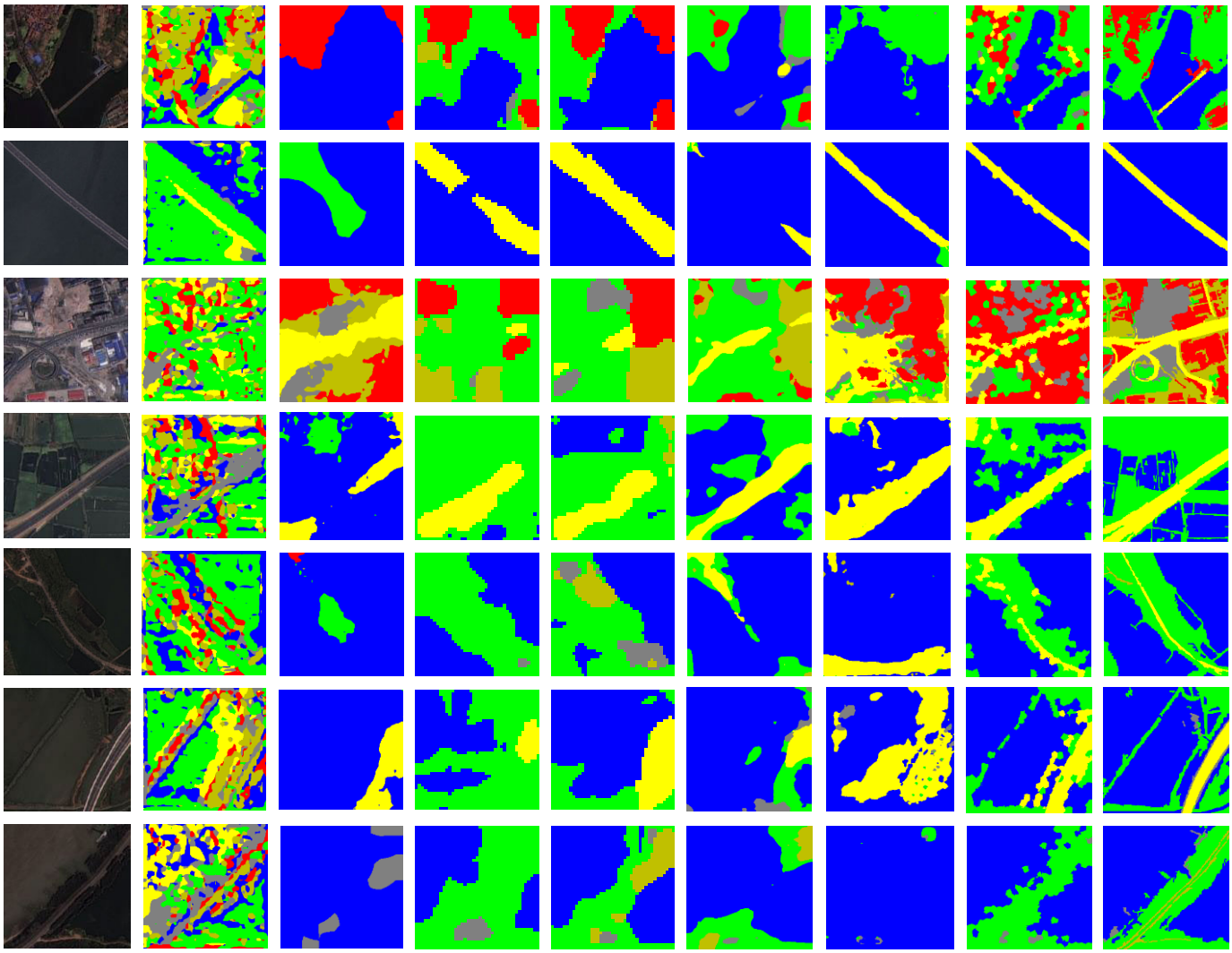


Fig. 8. Qualitative results of WSSS using different methods on WHDLD dataset. (a) Image. (b) MIL-FCN. (c) Grad-CAM. (d) SEC. (e) DSRG. (f) PSA. (g) SEAM. (h) Proposed. (i) Ground truth.

TABLE V
PER-CLASS RESULTS (%) OF IoU ON DLRSD DATASET

Method	aero	bare	build	cars	chap	court	dock	field	grass	mob	pave	sand	sea	ship	tanks	trees	water
MIL-FCN	0	4.15	12.86	2.80	0.01	0.01	0.01	0	9.33	0.04	23.46	1.13	4.76	0.16	0.30	19.44	0.50
Grad-CAM	14.61	3.19	6.33	17.60	42.73	54.38	17.36	93.47	37.58	**34.66**	43.87	26.55	36.94	8.77	**52.61**	44.45	31.12
SEC	11.47	30.25	38.25	6.98	21.68	0	11.07	97.13	35.33	25.41	39.89	26.75	60.86	15.46	48.33	56.63	36.44
DSRG	**15.31**	24.03	33.82	0.01	24.47	58.12	0	96.88	39.81	8.56	46.29	28.95	53.15	0.16	44.33	55.06	48.06
PSA	0.01	30.26	45.01	29.32	35.07	52.05	10.16	97.46	50.70	17.20	55.49	38.41	47.67	20.51	42.94	**63.04**	34.28
SEAM	0.02	28.72	**46.39**	**30.33**	39.95	57.43	27.21	88.37	**53.46**	32.83	**61.03**	33.93	**61.68**	3.06	38.42	40.03	36.77
proposed	0.08	**36.81**	40.56	16.07	**45.22**	**69.65**	0	**98.88**	50.63	24.55	50.06	**40.98**	58.38	**52.07**	51.32	59.01	**72.90**

process because they spend more time refining the edges of the initial segmentation results. Besides, compared with PSA and SEAM, our proposed approach not only provides faster training and testing processes but obtains more promising results, which proves that our method has advantages in practical application.

F. Parameters Setting

In this section, we set up experiments about the impact of superpixel patch with different sizes on the results because the processing unit in the GCSM is superpixel patch. For DLRSD dataset, we set the patch sizes to 8, 10, 12, 14, 16, 18, 20, 22, and 24. Besides, for WHDLD dataset,

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TABLE VI
PER-CLASS RESULTS (%) OF OA AND IoU ON WHDLD DATASET

Method	building		road		pavement		vegetation		bare soil		water	
	OA	IoU	OA	IoU	OA	IoU	OA	IoU	OA	IoU	OA	IoU
MIL-FCN	7.61	3.75	12.18	2.98	13.57	6.53	26.78	17.18	10.47	2.96	29.91	24.59
Grad-CAM	**47.48**	**25.96**	**64.46**	21.44	42.76	22.34	4.52	4.43	39.91	21.66	**96.74**	46.30
SEC	21.68	14.96	21.84	15.98	14.46	10.06	**81.44**	50.73	1.28	1.23	71.58	67.18
DSRG	22.10	13.39	30.95	20.32	18.64	11.40	66.81	45.13	23.55	15.39	77.95	67.16
PSA	24.59	17.82	40.25	**26.92**	27.63	17.38	49.67	37.19	40.90	20.89	92.08	58.77
SEAM	78.15	13.27	20.36	11.63	11.51	9.36	3.26	3.07	11.60	9.25	57.99	42.53
proposed	44.57	22.21	41.67	17.52	0	0	76.78	**60.99**	**44.05**	**21.96**	86.28	**76.01**

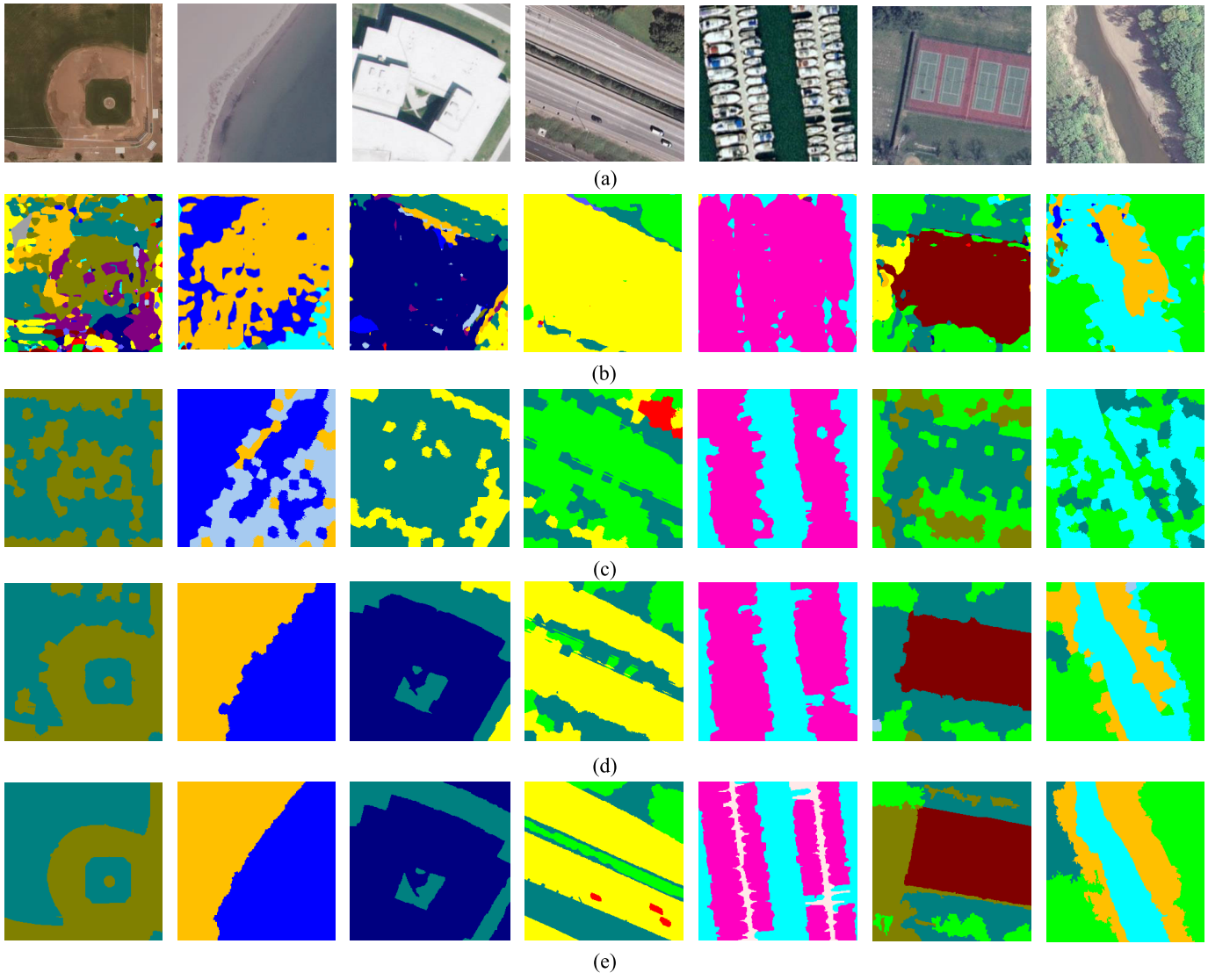


Fig. 9. Qualitative results of ablation study on DLRSD dataset. (a) Image. (b) Discriminative loss. (c) Similarity loss. (d) Discriminative loss + similarity loss. (e) Ground truth.

we set the sizes to 6, 8, 10, 12, 14, 16, 18, 20, and 22. The results with different patch sizes are shown in Fig. 11. Also, we show the inference speed (i.e., the number of images processed per second) for different sizes in Fig. 12.

The general trend of experimental results shows that with the increase of patch size, the quantitative performance of the method increases initially and decreases afterward, and we can obtain the best segmentation results when patch size is in a specific range. In addition, the increase in patch size will lead

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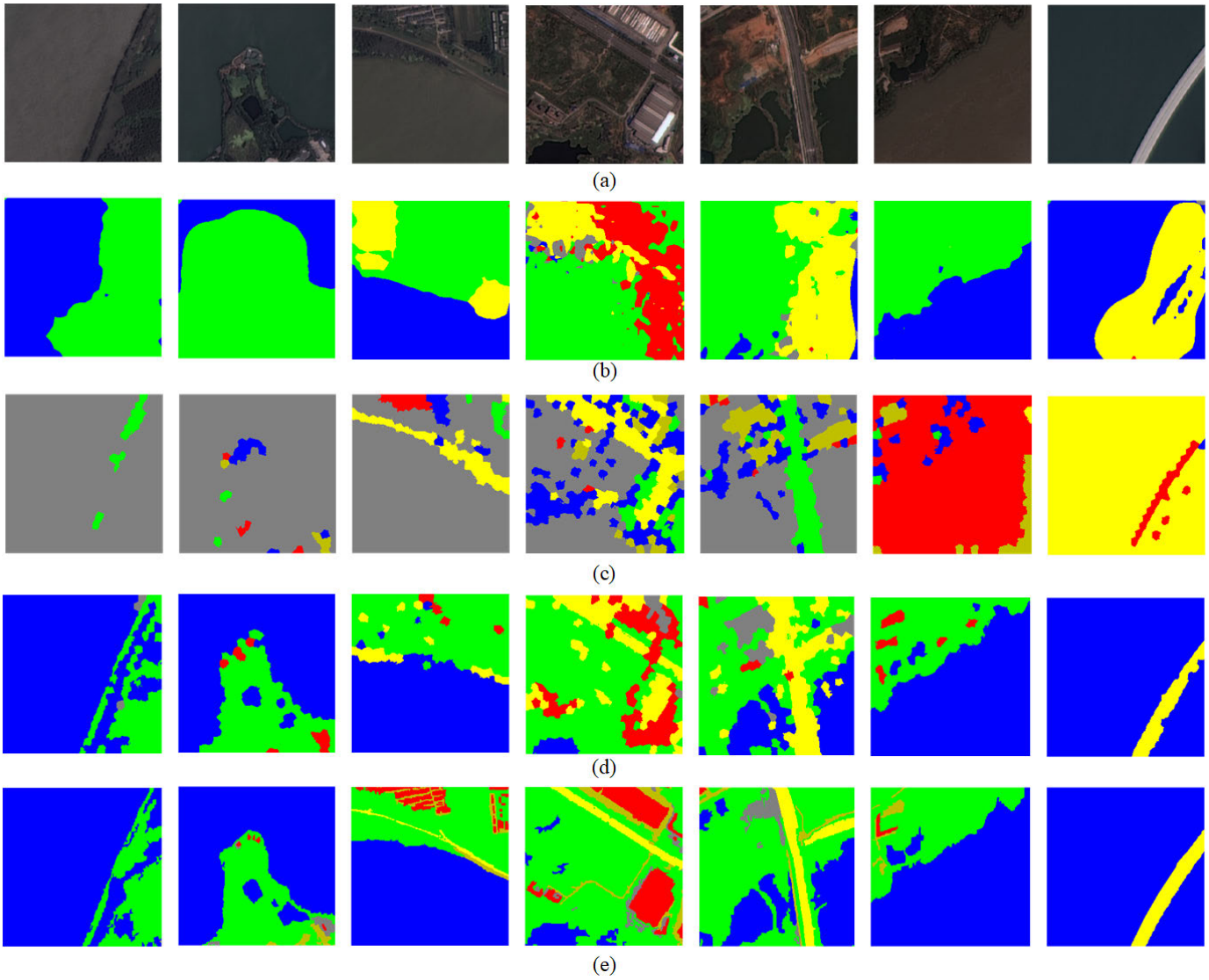


Fig. 10. Qualitative results of ablation study on WHDLD dataset. (a) Image. (b) Discriminative loss. (c) Similarity loss. (d) Discriminative loss + similarity loss. (e) Ground truth.

TABLE VII
EXPERIMENTAL RESULTS (%) OF ABLATION STUDY ON TWO DATASETS

Dataset	$\mathcal{L}_d$	$\mathcal{L}_s$	OA	recall	mIoU	fwIoU
DLRSD	√		57.10	57.58	34.18	41.05
		√	33.20	35.85	18.30	19.71
	√	√	64.28	62.43	45.58	50.42
WHDLD	√		51.95	38.00	23.61	36.05
		√	11.48	16.74	7.88	7.00
	√	√	63.99	48.89	33.11	50.02

to faster inference speed, because the larger the patch size is, the fewer the number of image segmentation superpixel patches and the less computational cost. To tradeoff between the segmentation accuracy and inference speed, we choose 16 and 12 as the patch sizes of DLRSD and WHDLD datasets, respectively. In fact, the selection of patch size is closely related to spatial resolution of remote sensing image scenes. When the value of patch size is too small, the difference between the features of ground regions belonging to the same category will be extended, and the computational consumption of the algorithm will be greatly improved. Besides, the edge information of surface regions, which is very important for the segmentation accuracy, will be lost if the size is too large. Therefore, we are supposed to determine the patch size that matches the spatial resolution of the used dataset, i.e., once the spatial resolution is large, the value of the patch size should be larger too and vice versa.

V. DISCUSSION

Deep learning has been widely applied in many tasks of computer vision. However, as a data-driven method, it is easy to fall into overfitting when training data are insufficient, leading to a decline in the generalization ability of the model. In addition, the model is already overfitted when using VGG16 to train the classification network, because the dataset contains only thousands of samples, so obviously, the number is very

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TABLE VIII
TIME–COST RESULTS (%) OF DIFFERENT METHODS ON TWO DATASETS

Dataset	Method	Training time (h)	Testing time (s)	Inference speed
DLRSD	MIL-FCN	0.7	34.1	61.6
	Grad-CAM	**0.6**	38.5	54.5
	SEC	3.2	**28.9**	**72.7**
	DSRG	3.1	50.7	41.4
	PSA	2.0	1555.0	1.4
	SEAM	1.5	975.3	2.2
	proposed	1.2	458.4	4.5
WHDLD	MIL-FCN	**1.4**	23.6	62.8
	Grad-CAM	1.5	**23.0**	**64.4**
	SEC	5.1	33.9	43.7
	DSRG	4.5	34.2	43.3
	PSA	4.4	1090.0	1.4
	SEAM	4.1	1071.8	1.4
	proposed	3.0	436.5	3.3

small. Therefore, we use VGG16 network with fewer layers rather than ResNet series networks [74], as the backbone to extract high-level semantic feature in the encoder. At the same time, our work aims to come up with a new base framework, so we used the basic network to validate our ideas.

Our proposed method fails for some classes due to the mechanism of the proposed SMG designed to extract saliency maps. The SMG learns features of the salient region of each class from the whole dataset. For example, for the samples containing freeway scene in the DLRSD dataset, their significant class is "road," and we extract the salient region by maximizing the energy of the "road" class in the output. Also, the region of "road" class in the freeway scene often coexists with the "grass" or "trees" classes, which are significant region classes in the golf course and forest scenes, respectively. Similarly, the features of "grass" or "trees" can be extracted by maximizing energy in the same way from their own samples. Thus, our method is capable of learning the feature of the most prominent classes in the image from the whole dataset. However, for some classes with a very small number of pixels and tiny shape, such as "dock" class in DLRSD dataset and "pavement" class in WHDLD dataset, the proportions of their pixels in images are so little that they are not significant classes in any image. More specifically, "dock" class only appears in the harbor scene with extremely few pixels on DLRSD dataset, so the significant class in the harbor scene is "water" or "ship." Besides, "pavement" class has the least proportion in each image on WHDLD dataset. Therefore, our method cannot learn their features and segment out regions of these classes, which is a limitation of our proposed method.

We would like to point out that there are two ways in which our work can be improved in the future. The first one is that we could improve our method for a more appropriate similarity measure method to design the loss function in the GSCM module. The similarity measure method we used is too basic to fully consider complex situations, so limiting its application. We will consider and measure the distance between different superpixel patches from a more comprehensive perspective instead of only using Euclidean distance in RGB space. Second, we could make our model learn the features of categories with very few proportions of pixels by adding more prior knowledge to address the class-imbalance problem.

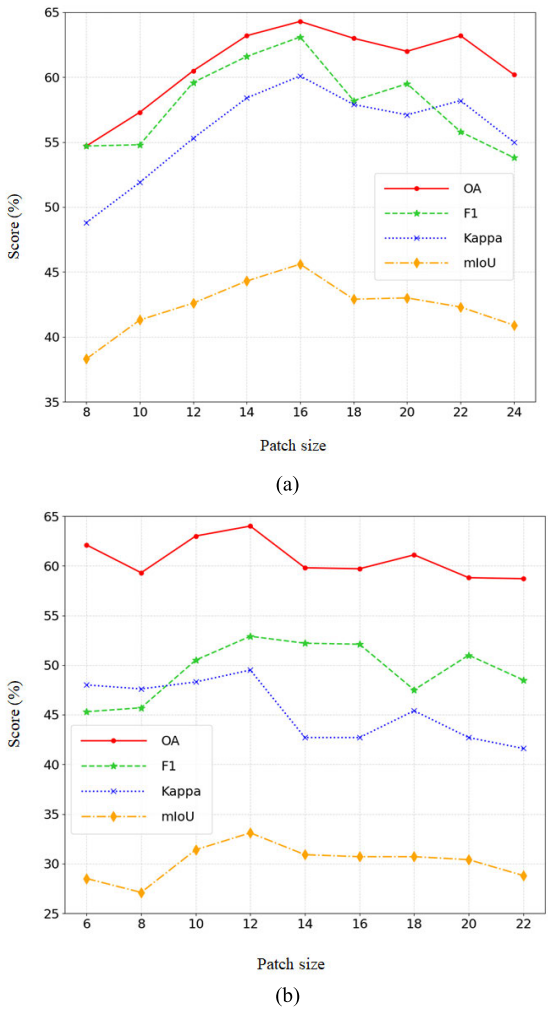


Fig. 11. Experimental results under different patch sizes on two datasets. (a) DLRSD dataset. (b) WHDLD dataset.

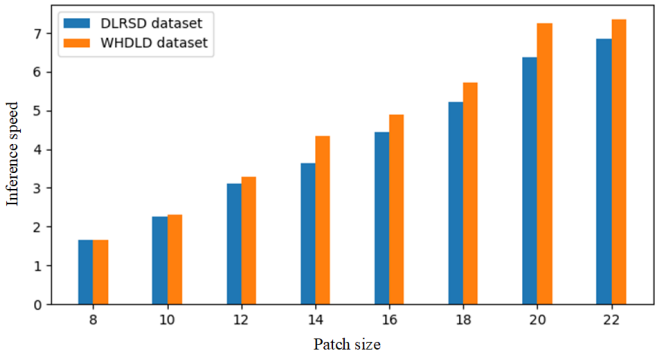


Fig. 12. Inference speed of the proposed method under different patch sizes.

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VI. Conclusion

In this article, we propose a novel framework for WSSS from remote sensing images using only image-level annotations. Different from other methods, we propose a saliency map generation method to provide more accurate location information, which outperforms the CAM and assigns semantic labels for the salient region of each class in the image. In addition, in order to make the segmentation results have better edges, we oversegment the input image and output semantic map into superpixel patches and propose a global similarity measure method based on the consistency between spatial and semantic similarity. Combined with the extracted salient region, we assign corresponding labels to each patch according to the saliency and similarity. Experimental results demonstrate that our method is superior to other WSSS methods and has both at least a 3% improvement in mIoU on two datasets.

Nowadays, few works focus on WSSS for remote sensing images, and our work is an early attempt of it. In the future, we will conduct further research on similarity measure and class-imbalance problem from a more comprehensive perspective to further improve the performance of the method. Moreover, we further believe that our method can be adapted to other tasks and other types of datasets.

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